

|                                                           |                                      |
|-----------------------------------------------------------|--------------------------------------|
| Program : <b>Diploma in Computer Hardware Engineering</b> |                                      |
| Course Code : <b>5152A</b>                                | Course Title: <b>Web Programming</b> |
| Semester : <b>5</b>                                       | Credits: <b>4</b>                    |
| Course Category: <b>Program Elective</b>                  |                                      |
| Periods per week: <b>4 (L:4 T:0 P:0)</b>                  | Periods per semester: <b>60</b>      |

### Course Objectives:

- To impart basics of the internet and HTML.
- Understand Cascading style sheets and scripting languages.
- To give an introduction to develop websites with database support.

### Course Prerequisites:

| Topic                                                  | Course Code | Course Name                    | Semester |
|--------------------------------------------------------|-------------|--------------------------------|----------|
| Basic Knowledge of IT skills and basic problem solving |             | Introduction to IT systems Lab | 1        |
| Object Oriented Programming concepts                   |             | Object Oriented Programming    | 4        |

### Course Outcomes:

On completion of the course student will be able to:

| CO <sub>n</sub> | Description                                                 | Duration (Hours) | Cognitive Level |
|-----------------|-------------------------------------------------------------|------------------|-----------------|
| CO1             | Demonstrate the fundamentals of World Wide Web and HTML.    | 14               | Understanding   |
| CO2             | Demonstrate client-side scripting using JavaScript and CSS. | 20               | Understanding   |
| CO3             | Demonstrate server-side scripting using PHP.                | 14               | Understanding   |
| CO4             | Build websites with database access using PHP.              | 10               | Applying        |
|                 | Series Test                                                 | 2                |                 |

**CO – PO Mapping**

| <b>Course Outcomes</b> | <b>PO 1</b> | <b>PO 2</b> | <b>PO 3</b> | <b>PO 4</b> | <b>PO 5</b> | <b>PO 6</b> | <b>PO 7</b> |
|------------------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|
| <b>CO1</b>             | 3           |             |             |             |             |             |             |
| <b>CO2</b>             | 3           |             |             |             |             |             |             |
| <b>CO3</b>             | 3           |             |             |             |             |             |             |
| <b>CO4</b>             | 3           |             |             |             |             |             |             |

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

**Course Outline:**

|                                                                                                                                                                                                                                                            | <b>Description</b>                                                                | <b>Hours</b> | <b>Cognitive Level</b> |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|--------------|------------------------|
| <b>CO1</b>                                                                                                                                                                                                                                                 | Demonstrate the fundamentals of World Wide Web and HTML.                          |              |                        |
| M1.01                                                                                                                                                                                                                                                      | Define world wide web and HTML.                                                   | 2            | Remembering            |
| M1.02                                                                                                                                                                                                                                                      | Demonstrate basic & formatting tags in HTML                                       | 4            | Understanding          |
| M1.03                                                                                                                                                                                                                                                      | Illustrate link and table using HTML tags                                         | 4            | Understanding          |
| M1.04                                                                                                                                                                                                                                                      | Build html forms using various controls                                           | 4            | Applying               |
| <b>Contents:</b> Define Internet, IPV, WWW & Domain names -web browsers, web servers-search engines & URL-HTML- Basic & formatting tags- HTML lists-Hyperlink and anchor tags-Frames & Tables- Embedding multimedia -Designing form with various controls. |                                                                                   |              |                        |
| <b>CO2</b>                                                                                                                                                                                                                                                 | Demonstrate client-side scripting using JavaScript and CSS                        |              |                        |
| M2.01                                                                                                                                                                                                                                                      | Explain cascading style sheets.                                                   | 4            | Understanding          |
| M2.02                                                                                                                                                                                                                                                      | Show relevance of client-side scripting and apply JavaScript in HTML page design. | 6            | Understanding          |
| M2.03                                                                                                                                                                                                                                                      | Demonstrate event handling in JavaScript                                          | 6            | Understanding          |
| M2.04                                                                                                                                                                                                                                                      | Make use of JavaScript in data validation.                                        | 4            | Applying               |
|                                                                                                                                                                                                                                                            | Series Test-I                                                                     | 1            |                        |

**Contents:** CSS-various methods for embedding CSS-inline-internal & external-CSS selectors-properties-font, color, background, list, link, text, border in web pages. Need for client-side scripting-use of JavaScript in HTML -programming elements in JavaScript-DOM-Event handling in java script-data validation using JavaScript

|            |                                                    |   |               |
|------------|----------------------------------------------------|---|---------------|
| <b>CO3</b> | <b>Demonstrate server-side scripting using PHP</b> |   |               |
| M3.01      | Compare client/server-side scripting               | 1 | Understanding |
| M3.02      | Explain programming elements in PHP.               | 6 | Understanding |
| M3.03      | Illustrate form handling using PHP                 | 4 | Understanding |
| M3.04      | Demonstrate the use of sessions and cookies        | 3 | Understanding |

**Contents:** Comparison of client and server-side scripting -popular web servers-Features of PHP-embedding php scripts in a web page-PHP language elements-PHP arrays and functions - Retrieving and processing form data using PHP - setting and retrieving cookies - Session handling in PHP-PDF generation in PHP.

|            |                                                           |   |               |
|------------|-----------------------------------------------------------|---|---------------|
| <b>CO4</b> | <b>Build websites with database access using PHP.</b>     |   |               |
| M4.01      | Explain how to establish a connection with MySQL.         | 2 | Understanding |
| M4.02      | Illustrate SQL statement execution using PHP.             | 4 | Understanding |
| M4.03      | Illustrate file uploading procedure                       | 2 | Understanding |
| M4.04      | Show Domain Name Registration & server/cloud space hiring | 1 | Understanding |
| M4.05      | Explain Website Hosting and its types                     | 1 | Understanding |
|            | Series Test – II                                          | 1 |               |

**Contents:** Uses of Databases in Web Applications – Database Connection using PHP – SQL Statements Execution – Select, Insert, Update & Delete Operations in MySQL Database using PHP-Design and develop simple complete database driven multi tiered web applications using HTML, CSS, JavaScript, PHP & MySQL- File uploading procedure - Domain Name Registration – Server/cloud space hiring– Website Hosting – types of website hosting.

#### **Text / Reference:**

| <b>T/R</b> | <b>Book Title/Author</b>                                                                       |
|------------|------------------------------------------------------------------------------------------------|
| T1         | Robert W Sebesta, Programming with World Wide Web, 7th ed., Pearson Education, New Delhi, 2009 |
| R1         | Welling, PHP and MySQL Web Development, Fourth Edition, Pearson                                |

|    |                                                                                                              |
|----|--------------------------------------------------------------------------------------------------------------|
|    | Education                                                                                                    |
| R2 | Sam's teach Yourself PHP in 24 hours – Zandstra – (Pearson Education)-Third Edition                          |
| R3 | Beginning Web Programming with HTML, XHTML, CSS & JavaScript – John Duckett (Wiley DreamTech)-Second Edition |

### Online Resources:

| Sl.No | Website Link                                                                                                                                                        |
|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1     | <a href="http://www.w3schools.com">http://www.w3schools.com</a>                                                                                                     |
| 2     | <a href="http://php.net">http://php.net</a>                                                                                                                         |
| 3     | <a href="http://www.mysql.com">http://www.mysql.com</a>                                                                                                             |
| 4     | <a href="https://www.tutorialspoint.com/internet_technologies/websites_overview.htm">https://www.tutorialspoint.com/internet_technologies/websites_overview.htm</a> |